Perishable Inventory Control with Shelf-Life Constraints: A Reinforcement Learning Approach

Nilay PURAYAR & Vishakh SHAH

Introduction & Motivation

- Managing perishable goods (food, pharmaceuticals, blood products) poses a critical supply-chain challenge where **unsold items expire** and **unmet demand causes stockouts**.
- Ordering too much ⇒ **waste** (expired units, financial loss).
  Ordering too little ⇒ **stockouts** (lost sales, penalty costs).
- The optimal ordering policy must balance these **competing objectives** under **stochastic demand** - a sequential decision problem with **delayed consequences**.
- Classical heuristics (e.g., order-up-to) ignore the **age-structure** of inventory; RL methods can discover policies that condition on the full age profile.

Context: Demand & Perishability

- **Demand:** $d_t \sim \text{Poisson}(\lambda)$, default $\lambda = 6$ units/day, truncated for DP.
- **Shelf-life:** D days (default $D = 3$); units age through D age-buckets.
- **Selling rule:** FEFO (First-Expire-First-Out) — oldest items sold first.
- **Episode horizon:** $T = 60$ days.

Problem Formulation (MDP)

State Space

$$S_t = (n_1, n_2, \ldots, n_D), \quad n_i \in \{0, \ldots, N_{\max}\}$$

n_i = units with i days of remaining shelf-life.
State-space size: $(N_{\max} + 1)^D$.

Action Space

$$a_t \in \{0, 1, \ldots, A_{\max}\}$$

Transition (one step)

1. Receive order: $n_D \leftarrow \min(n_D + a_t,\ N_{\max})$
2. Sample demand: $d \sim \text{Poisson}(\lambda)$
3. Sell FEFO: sell from oldest bucket first
4. Waste: leftover in bucket 1 expires
5. Age: $n_i \leftarrow n_{i+1}, \quad n_D \leftarrow 0$

Reward Function

$$R_t = \underbrace{p \cdot \text{sold}}_{\text{revenue}} - \underbrace{c \cdot a_t}_{\text{order cost}} - \underbrace{w \cdot \text{waste}}_{\text{waste penalty}} - \underbrace{s \cdot \text{stockout}}_{\text{stockout penalty}}$$

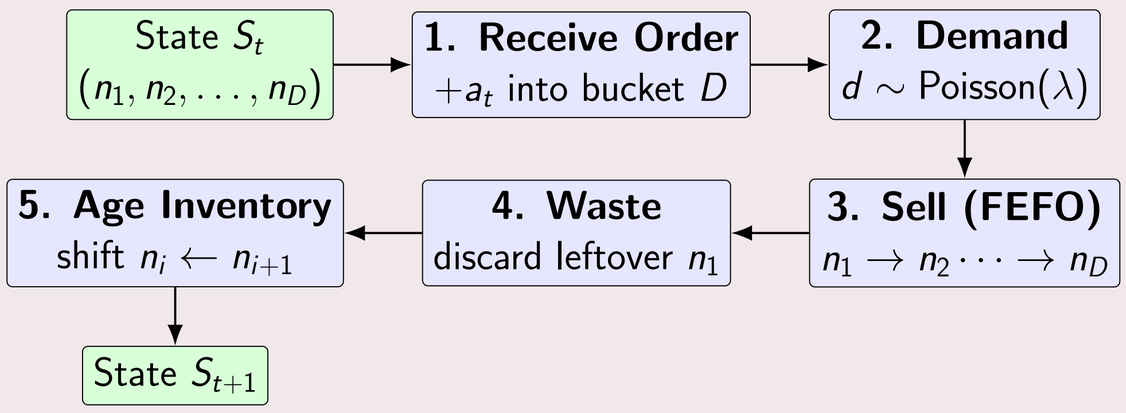
$$p = 10 \quad c = 4 \quad w = 6 \quad s = 8$$

Methodology

- Custom **Gymnasium-style** simulation environment with `reset()`/`step()` API.
- Pure-function `simulate_step()` for model-based DP; stochastic `step()` for model-free agents.
- Fixed-seed evaluation protocol (seed 123, 100 episodes) for fair comparison.

Agent Architectures

Algorithm	Type	Update
DP – Value Iter.	Model-based	$V(s) \leftarrow \max_a \sum_d P(d)[R + \gamma V(s')]$
Q-Learning	Off-policy TD	$Q \leftarrow Q + \alpha[r + \gamma \max_{a'} Q' - Q]$
SARSA	On-policy TD	$Q \leftarrow Q + \alpha[r + \gamma Q(s', a') - Q]$
Monte Carlo	First-visit	$Q \leftarrow Q + \alpha[G - Q]$
Linear FA	Semi-grad. SARSA	$\mathbf{w}_a \leftarrow \mathbf{w}_a + \alpha\delta\phi(s)$

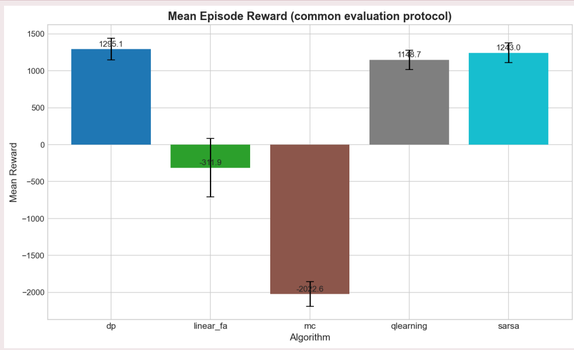


Intra-day state transition sequence representing the FEFO environment logic.

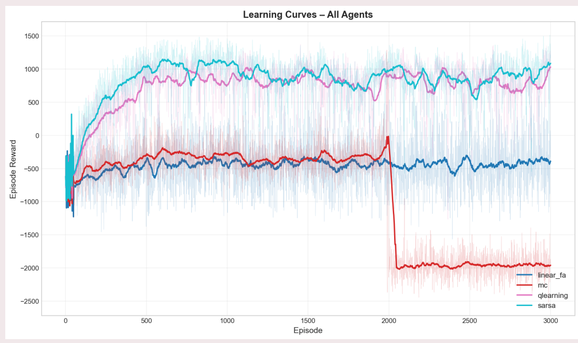
Experimental Setup

Parameter	Value
Training episodes	3 000
Episode horizon T	60 days
Learning rate α	0.1
Discount factor γ	0.99
ε start / min / decay	1.0 / 0.05 / 0.995
Demand mean λ	6.0 (Poisson)
Shelf-life D	3
$A_{\max}$ / $N_{\max}$	5 / 10
Seed (train / eval)	42 / 123
DP threshold θ	10^{-6}

Results Overview



Mean eval. reward ($\pm\sigma$) across agents.



Smoothed learning curves.

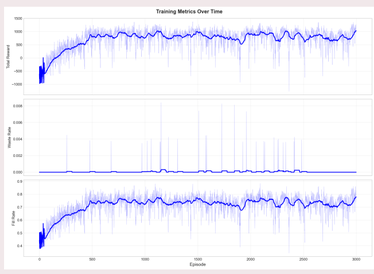
Quantitative Comparison

Agent	Return (mean $\pm\sigma$)	Waste (%)	Stockout (%)	Fill (%)
DP (VI)	1295.1 ± 145.9	0.24	16.5	83.5
Q-Learning	1148.7 ± 129.9	0.04	19.6	80.4
SARSA	1243.0 ± 132.9	0.07	17.7	82.3
Monte Carlo	-2022.6 ± 165.9	0.00	83.2	16.8
Linear FA	-311.9 ± 397.3	0.18	48.6	51.4

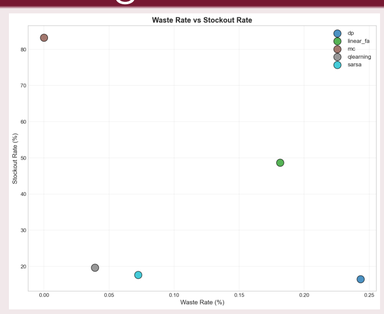
Key Findings

- **DP** provides the theoretical upper bound but is intractable for large D or $N_{\max}$ (state explosion $(N_{\max}+1)^D$).
- **Q-Learning & SARSA** achieve near-optimal performance via tabular TD, converging within ∼2 000 episodes.
- **SARSA** (on-policy) is more conservative: lower waste, slightly higher stockout — reflecting the exploration–exploitation trade-off in ordering.
- **Monte Carlo** converges slower due to high variance over the 60-step horizon.

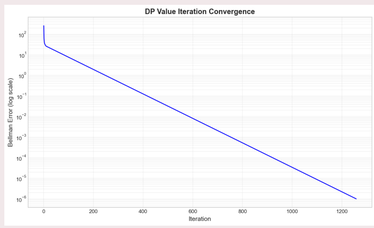
Learning Dynamics & Convergence



Q-Learning metrics.



Waste vs. Stockout.



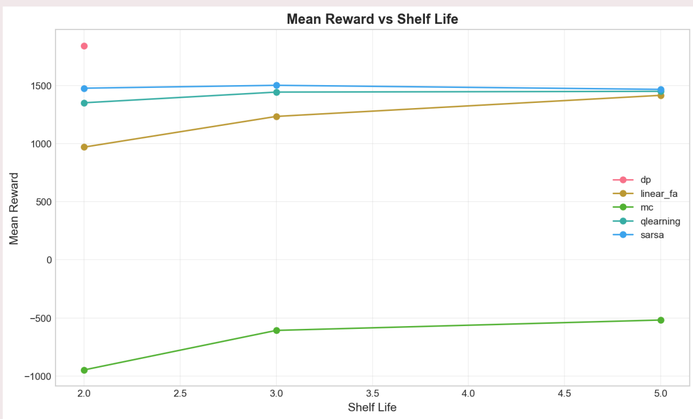
DP Bellman error (log).

Sensitivity Analysis

Swept parameters (3 seeds × 1 000 episodes each):

Parameter	Values
Shelf-life D	2, 3, 5
Demand mean λ	3.0, 6.0, 9.0
Waste penalty w	2.0, 6.0, 10.0
Stockout penalty s	4.0, 8.0, 12.0

- Shorter shelf-life (D=2) ⇒ significantly higher waste rates.
- Higher λ stresses all agents ⇒ increased waste *and* stockout.
- Increasing w drives under-ordering (less waste, more stockout).
- Increasing s drives over-ordering (less stockout, more waste).



Mean reward vs. shelf-life D (per agent).

Conclusion & Future Work

Conclusion

- **DP** yields the optimal baseline but scales as $\mathcal{O}\big((N_{\max}+1)^D\big)$.
- **Tabular TD** (Q-Learning, SARSA) reaches near-optimal performance without a model.
- **Linear FA** bridges tabular and deep RL with only D+3 hand-crafted features.
- The **waste–stockout trade-off** is the defining tension; on-policy methods favour safety.

Future Work

1. **Deep RL (DQN / PPO):** Neural function approximators for realistic $D \geq 7$ and continuous order quantities.
2. **Non-stationary & multi-product demand:** Time-varying distributions (seasonal), shared warehouse capacity — closer to real retail.